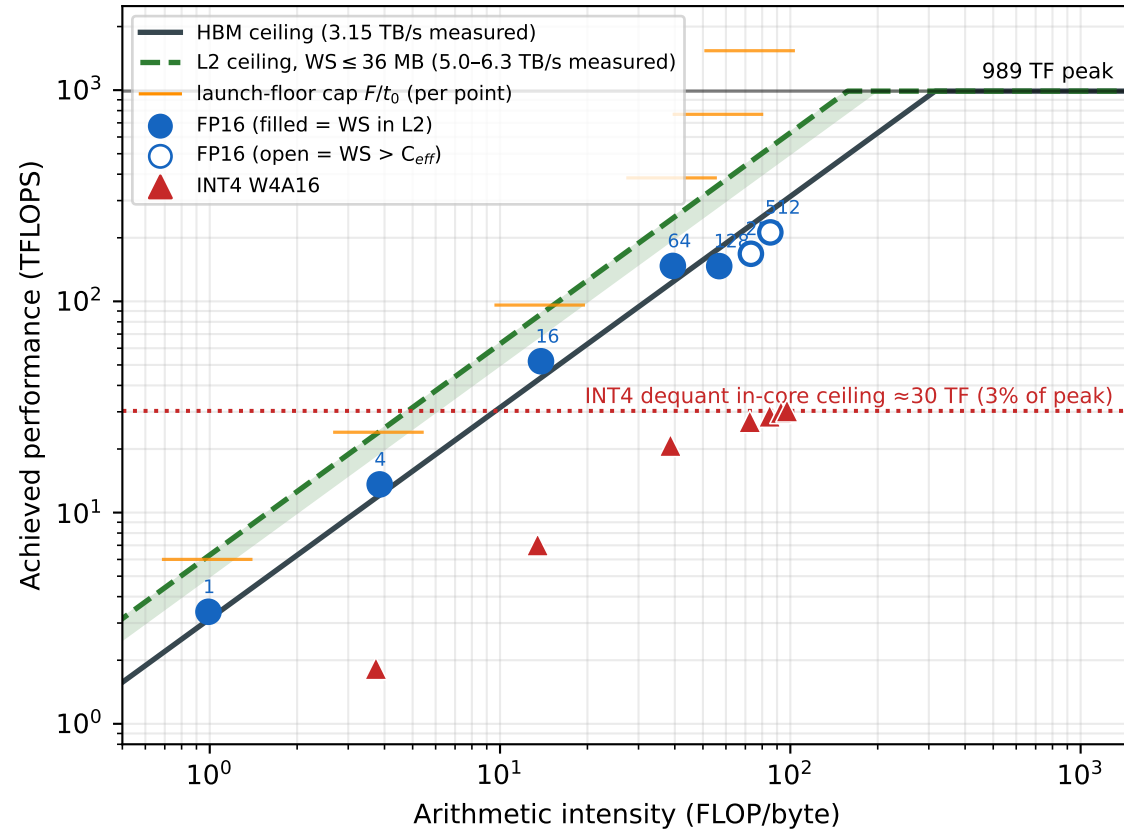
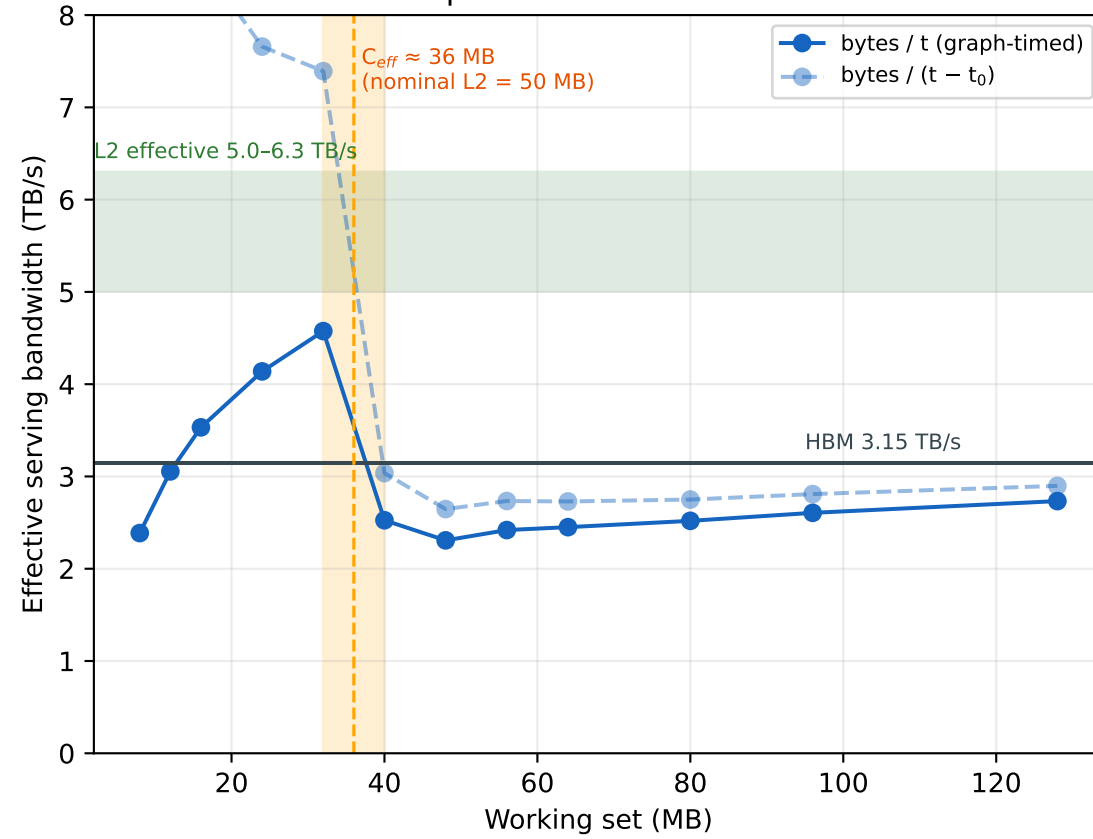


Cache-Aware Roofline (measured) — MLA reconstruction BMM1 on NVIDIA H100 80GB HBM3
 $BW_{HBM}=3.15$ TB/s, $BW_{L2}^{eff}=6.3$ TB/s (GEMM) / 5.0 TB/s (reduction), $C_{eff}=36$ MB, $t_0=2.8$ μ s graph / 15.4 μ s eager, $R_{dq}=0.50$ TB/s

A — CARM with measured ceilings + per-launch floor caps
 (graph-timed operating points, labels = batch size)



B — Capacity-gated bandwidth $BW(WS)$:
 the step function the model uses



C — Model validation across batch sizes
 FP16 MAPE 10% · INT4 MAPE 18% (dotted = $\pm 25\%$)

